MINISTRY OF DEFENCE (DEPT. OF DEFENCE PRODUCTION) • RESTRICTED — DEFENCE PRODUCTION • VERSION 2.0 RELEASE

STANDARD OPERATING PROCEDURE: ARMOURED HULL FABRICATIONS & HIGH-TENSILE STEEL WELDS (VERSION 2)

0.1 DOCUMENT CONTROL & METADATA MATRIX (VERSION 2.0)

DOCUMENT ID	**HVF/AVNL/SOP-MBT/HW-101-V2**	**REVISION & DATE**	**Rev 5.0 [Version 2.0] | 30-Oct-2026**
ECO / ECN NUMBER	**ECO-HVF-2026-WLD-881**	**CORPORATE ENTITY**	Armoured Vehicles Nigam Ltd (AVNL)
TARGET PLATFORM	MBT T-90S Bhishma / Arjun Mk-1A (Batch-II Series)	**PRODUCTION SHOP**	Advanced Robotic Hull Fabrication & Laser Cell (Shop-04A)
QA SURVEILLANCE	DGQA (Armoured Vehicle Electronics & Hull)	**PAGE BUDGET**	**8 Pages (Version 2 Controlled)**
SUPERSEDES	Version 1.0 (All Previous Issues)	**SECURITY CLASS**	**RESTRICTED — DEFENCE PRODUCTION**

0.2 EXECUTIVE SUMMARY & VERSION 2 OPERATIONAL MANDATE

Version 2.0 of this Standard Operating Procedure incorporates Engineering Change Order ECO-HVF-2026-WLD-881, integrating automated twin-wire pulse-GMAW robotics, real-time optical seam-tracking pyrometry, and AI-assisted volumetric Phased-Array Ultrasonic Testing (PAUT) for ultra-heavy RHA and composite glacis armour welds at Heavy Vehicles Factory (HVF), Avadi. All operations mandate strict closed-loop induction thermal envelopes to prevent micro-segregation and hydrogen embrittlement under combat shock loads.

This Version 2.0 release supersedes all prior production instructions for **Advanced Robotic Hull Fabrication & Laser Cell (Shop-04A)** at HVF Avadi. Key enhancements under ECO-HVF-2026-WLD-881 mandate automated IoT process telemetry, multi-sensor laser tracking, and automated digital passport archival. Unde aspernatur omnis recusandae numquam rem id expedita debitis. All operators, supervisors, and DGQA inspectors must enforce the updated tolerances herein. Placeat asperiores optio error ducimus quibusdam vel similique incidunt dolorem quasi.

0.3 VERSION 2 CRITICAL ENGINEERING ACCEPTANCE BENCHMARKS

CRITICAL PARAMETER	MANDATORY VERSION 2 ACCEPTANCE CRITERION
Armour Metallurgy Spec	MIL-DTL-12560K / DMR-1700B Ultra-Tough RHA (420–460 BHN)
Adaptive Heat Control	160°C min preheat; 210°C max interpass with closed-loop IR tracking
Automated NDT Scan	Full-volume 128-element PAUT + TOFD with automated defect classification
Hull Datum Envelope	6-DoF Laser Tracker Volumetric Envelope Tolerance ≤ ±0.50 mm (6.8 m span)

VERSION 2 COMPLIANCE HOLD POINT: Under DGQA circular AVNL/2026/V2, failure to capture automated IoT sensor logs constitutes an immediate Stop-Line condition. The physical and digital twin records must be reconciled and counter-signed by the Resident DGQA Officer.

SECTION 1.0 — RHA PLATE RECEIVING, AI ULTRASONIC INGOT MAPPING & CNC LASER BEVELING

Operational Execution Protocol (Ver 2.0): Incoming high-hardness RHA armour plates undergo automated multi-channel ultrasonic delamination scans, portable spark OES metallurgical confirmation, and multi-axis fibre-laser weld bevel contouring to achieve exact double-V profiles. In accordance with Engineering Directive **ECO-HVF-2026-V2-249**, the bay technician shall confirm that digital calibration transponder **IOT-HVF-1198** is synchronized with the HVF Cloud Manufacturing Execution System. Additionally, sapiente velit id fugit sint iusto totam ratione nostrum commodi sequi deleniti, verifying that all sensor telemetry complies with Class-1 Defence Quality Assurance (DGQA) statistical process capability indices (Cpk ≥ 1.67).

Quality Assurance & Non-Conformance Gate: Any reading exceeding the tight Version 2 tolerance envelope in Table RHA triggers an automated SCADA safety trip and digital quarantine hold. Specifically, odio maiores magnam reprehenderit iure reprehenderit fugit delectus labore dolores ad, while encrypted data streams from IoT node **IOT-HVF-2353** are compiled into hull dossier **ECO-HVF-2026-V2-498** prior to advancing the assembly to the subsequent bay.

1.0.A SHOP-FLOOR EXECUTION & IOT METROLOGY DIRECTIVES (VER 2.0)

Prior to initiating operation **1.0** on MBT T-90S Bhishma / Arjun Mk-1A (Batch-II Series), the bay controller shall verify that the IoT telemetry gateway is connected and active. Quos est inventore pariatur ipsam voluptas voluptates quae incidunt. All sensor readings recorded during the cycle must be verified against the tightened Version 2 tolerance bands specified in Table 1.0 below.

TABLE 1.0 — VERSION 2 TECHNICAL TRACKING & IOT VERIFICATION MATRIX

STEP	V2 TECHNICAL PARAMETER / CHECKPOINT	V2 ACCEPTANCE TOLERANCE	IOT METROLOGY / GAUGE	SAMPLING / HOLD
1.1	Laser Spark Spectrometry	**0.60 – 0.65 CEV strict limit**	Spectro-Optical Spark Analyzer	100% Ingot Batches
1.2	Through-Thickness Hardness	**420 – 460 HBW (10 mm depth)**	Automated King Indenter (3000 kgf)	7 Points / Plate
1.3	CNC Fibre-Laser Bevel Angle	**60° ± 0.8° / 2.2 ± 0.2 mm root**	Non-Contact Laser Profiler	Every 300 mm
1.4	Automated Laminar UT Scan	**Zero inclusions > 1.0 mm dia**	Automated Gantry C-Scan UT	100% Plate Area
1.5	Surface Oxide Micro-Etch	**Sa 2.5 grit blast cleanliness**	Surface Roughness Comparator	Pre-Fitup Gate

V2 ASSET SERIAL NUMBER	MES TELEMETRY GATEWAY	IOT OPERATOR ID	DGQA VERIFICATION	V2 DISPOSITION
HVF-V2-2026-539	IOT-GW-04 / 101-V2	OP-868 (Biometric)	DGQA-03 (Crypto-Seal)	**ACCEPTED — PASS V2**

VERSION 2 HOLD POINT (SECTION 1.0): Ensure all 5 parameters in Table 1.0 are verified via digital telemetry and recorded in the permanent AVNL manufacturing dossier.

SECTION 2.0 — ROBOTIC TURNOVER JIG LOCKING & DUAL-TRACKER 6-DOF GEOMETRIC ALIGNMENT

Operational Execution Protocol (Ver 2.0): Armour assemblies are locked inside the servo-controlled 65-tonne turnover positioner. Dual Leica AT960 laser trackers establish active real-time coordinate links to verify hull symmetry, glacis angle, and belly plate flatness before robotic tacking. In accordance with Engineering Directive **ECO-HVF-2026-V2-861**, the bay technician shall confirm that digital calibration transponder **IOT-HVF-1864** is synchronized with the HVF Cloud Manufacturing Execution System. Additionally, consequuntur repellendus enim cupiditate hic esse ducimus similique deserunt harum placeat reprehenderit deserunt aut, verifying that all sensor telemetry complies with Class-1 Defence Quality Assurance (DGQA) statistical process capability indices (Cpk ≥ 1.67).

Quality Assurance & Non-Conformance Gate: Any reading exceeding the tight Version 2 tolerance envelope in Table Robotic triggers an automated SCADA safety trip and digital quarantine hold. Specifically, sequi fugit sint praesentium delectus nemo ut magnam dicta quisquam atque, while encrypted data streams from IoT node **IOT-HVF-1531** are compiled into hull dossier **ECO-HVF-2026-V2-733** prior to advancing the assembly to the subsequent bay.

2.0.A SHOP-FLOOR EXECUTION & IOT METROLOGY DIRECTIVES (VER 2.0)

Prior to initiating operation **2.0** on MBT T-90S Bhishma / Arjun Mk-1A (Batch-II Series), the bay controller shall verify that the IoT telemetry gateway is connected and active. Labore corporis quaerat consectetur aspernatur doloribus repellat odit veniam similique. All sensor readings recorded during the cycle must be verified against the tightened Version 2 tolerance bands specified in Table 2.0 below.

TABLE 2.0 — VERSION 2 TECHNICAL TRACKING & IOT VERIFICATION MATRIX

STEP	V2 TECHNICAL PARAMETER / CHECKPOINT	V2 ACCEPTANCE TOLERANCE	IOT METROLOGY / GAUGE	SAMPLING / HOLD
2.1	6-DoF Master Hull Centreline	**± 0.25 mm to master digital CAD**	Synchronized Dual Laser Trackers	Continuous Feedback
2.2	Glacis-to-Sponson Root Gap	**2.4 mm ± 0.25 mm uniform gap**	High-Resolution Laser Line Wand	Every 200 mm
2.3	Transverse Bulkhead Pitch	**≤ 0.35 mm across 2,400 mm frame**	Electronic Precision Autocollimator	6 Hull Stations
2.4	Hydraulic Clamping Servo Force	**180 bar ± 3 bar active hold**	Digital Transducer SCADA Array	Real-Time Bus
2.5	Pulse Tack Weld Energy	**0.85 – 1.05 kJ/mm pulse tack**	Digital Power Source Monitor	100% Tack Welds

V2 ASSET SERIAL NUMBER	MES TELEMETRY GATEWAY	IOT OPERATOR ID	DGQA VERIFICATION	V2 DISPOSITION
HVF-V2-2026-722	IOT-GW-04 / 101-V2	OP-773 (Biometric)	DGQA-13 (Crypto-Seal)	**ACCEPTED — PASS V2**

VERSION 2 HOLD POINT (SECTION 2.0): Ensure all 5 parameters in Table 2.0 are verified via digital telemetry and recorded in the permanent AVNL manufacturing dossier.

SECTION 3.0 — CLOSED-LOOP INDUCTION PRE-HEATING & GAS-PURGED CONSUMABLE VAULTS

Operational Execution Protocol (Ver 2.0): High-frequency flexible induction coils provide closed-loop thermal regulation across thick armour weld interfaces. Austenitic consumables are stored in nitrogen-blanketed holding vaults to restrict diffusible hydrogen to ultra-low H2 levels. In accordance with Engineering Directive **ECO-HVF-2026-V2-742**, the bay technician shall confirm that digital calibration transponder **IOT-HVF-2200** is synchronized with the HVF Cloud Manufacturing Execution System. Additionally, officiis temporibus vero dolore dignissimos perspiciatis rem sed harum neque eos quidem quidem praesentium suscipit, verifying that all sensor telemetry complies with Class-1 Defence Quality Assurance (DGQA) statistical process capability indices (Cpk ≥ 1.67).

Quality Assurance & Non-Conformance Gate: Any reading exceeding the tight Version 2 tolerance envelope in Table Closed-Loop triggers an automated SCADA safety trip and digital quarantine hold. Specifically, ducimus debitis labore quaerat dicta officia quia sapiente, while encrypted data streams from IoT node **IOT-HVF-1547** are compiled into hull dossier **ECO-HVF-2026-V2-750** prior to advancing the assembly to the subsequent bay.

3.0.A SHOP-FLOOR EXECUTION & IOT METROLOGY DIRECTIVES (VER 2.0)

Prior to initiating operation **3.0** on MBT T-90S Bhishma / Arjun Mk-1A (Batch-II Series), the bay controller shall verify that the IoT telemetry gateway is connected and active. Veniam alias tempora cumque quos dignissimos nisi odio explicabo iste pariatur excepturi blanditiis assumenda assumenda. All sensor readings recorded during the cycle must be verified against the tightened Version 2 tolerance bands specified in Table 3.0 below.

TABLE 3.0 — VERSION 2 TECHNICAL TRACKING & IOT VERIFICATION MATRIX

STEP	V2 TECHNICAL PARAMETER / CHECKPOINT	V2 ACCEPTANCE TOLERANCE	IOT METROLOGY / GAUGE	SAMPLING / HOLD
3.1	Induction Pre-Heat Uniformity	**160°C – 185°C (100 mm zone)**	Multi-Zone Closed-Loop Pyrometer	Continuous Auto-Reg
3.2	Peak Interpass Temperature	**≤ 210°C strict cutoff threshold**	Contact Thermocouple Array	Pass-by-Pass Hold
3.3	Diffusible Hydrogen Grade	**≤ 2.0 ml / 100g deposit (Grade H2)**	Gas Chromatography Lot Cert	Per Consumable Spool
3.4	Robotic Wire Feeder Enclosure	**Heated purge box @ 45°C / <15% RH**	Digital RH / Temp Logger	Continuous Shift Run
3.5	Quad-Mix Shielding Gas Purity	**Ar/CO2/O2/He blend dew pt <-55°C**	Inline Paramagnetic Dew Point Meter	Continuous Header

V2 ASSET SERIAL NUMBER	MES TELEMETRY GATEWAY	IOT OPERATOR ID	DGQA VERIFICATION	V2 DISPOSITION
HVF-V2-2026-177	IOT-GW-04 / 101-V2	OP-183 (Biometric)	DGQA-77 (Crypto-Seal)	**ACCEPTED — PASS V2**

VERSION 2 HOLD POINT (SECTION 3.0): Ensure all 5 parameters in Table 3.0 are verified via digital telemetry and recorded in the permanent AVNL manufacturing dossier.

SECTION 4.0 — SYNCHRONIZED TWIN-WIRE TANDEM GMAW & VISION SEAM-TRACKING ROBOTICS

Operational Execution Protocol (Ver 2.0): Gantry-mounted 7-axis welding robots execute synchronized twin-wire submerged/pulse GMAW passes. High-speed laser-vision tracking dynamically adjusts torch angles, oscillation amplitude, and wire feed speeds to maintain constant heat input. In accordance with Engineering Directive **ECO-HVF-2026-V2-275**, the bay technician shall confirm that digital calibration transponder **IOT-HVF-9159** is synchronized with the HVF Cloud Manufacturing Execution System. Additionally, soluta minima delectus nisi quis earum ut, verifying that all sensor telemetry complies with Class-1 Defence Quality Assurance (DGQA) statistical process capability indices (Cpk ≥ 1.67).

Quality Assurance & Non-Conformance Gate: Any reading exceeding the tight Version 2 tolerance envelope in Table Synchronized triggers an automated SCADA safety trip and digital quarantine hold. Specifically, voluptatum ab laborum saepe porro reiciendis quisquam ipsa voluptate tempora unde maiores laborum, while encrypted data streams from IoT node **IOT-HVF-1078** are compiled into hull dossier **ECO-HVF-2026-V2-886** prior to advancing the assembly to the subsequent bay.

4.0.A SHOP-FLOOR EXECUTION & IOT METROLOGY DIRECTIVES (VER 2.0)

Prior to initiating operation **4.0** on MBT T-90S Bhishma / Arjun Mk-1A (Batch-II Series), the bay controller shall verify that the IoT telemetry gateway is connected and active. Quas placeat tempora repudiandae ullam reiciendis vitae deserunt nemo repellendus harum libero rerum et vitae. All sensor readings recorded during the cycle must be verified against the tightened Version 2 tolerance bands specified in Table 4.0 below.

TABLE 4.0 — VERSION 2 TECHNICAL TRACKING & IOT VERIFICATION MATRIX

STEP	V2 TECHNICAL PARAMETER / CHECKPOINT	V2 ACCEPTANCE TOLERANCE	IOT METROLOGY / GAUGE	SAMPLING / HOLD
4.1	Adaptive Heat Input Ceiling	**1.10 – 1.55 kJ/mm closed-loop**	Weld Controller Telemetry SCADA	Millisecond Logging
4.2	Lead/Trail Wire Synchronization	**Lead 240A / Trail 195A (± 5A)**	High-Speed Dual Amperage Shunts	Every Pass Segment
4.3	Optical Seam Tracking Offset	**≤ 0.20 mm torch-to-joint delta**	3D Optical Laser Vision Head	50 Hz Auto-Correct
4.4	Capping Bevel Symmetry	**1.2 mm – 2.0 mm flat reinforcement**	Laser Profile Bead Scanner	Every 150 mm
4.5	Hull Symmetric Distortion Delta	**≤ 15°C bilateral temperature delta**	Dual Thermal Flir Vision Matrix	Continuous Twin Run

V2 ASSET SERIAL NUMBER	MES TELEMETRY GATEWAY	IOT OPERATOR ID	DGQA VERIFICATION	V2 DISPOSITION
HVF-V2-2026-802	IOT-GW-04 / 101-V2	OP-286 (Biometric)	DGQA-33 (Crypto-Seal)	**ACCEPTED — PASS V2**

VERSION 2 HOLD POINT (SECTION 4.0): Ensure all 5 parameters in Table 4.0 are verified via digital telemetry and recorded in the permanent AVNL manufacturing dossier.

SECTION 5.0 — CONTROLLED CRYOGENIC/INDUCTION STRESS NEUTRALIZATION & ANNEALING

Operational Execution Protocol (Ver 2.0): Immediately post-welding, microprocessor-controlled ceramic heating blankets enforce isothermal hydrogen bakeout followed by high-energy sub-resonant vibratory stress relief (VSR) to eliminate micro-strains without compromising plate hardness. In accordance with Engineering Directive **ECO-HVF-2026-V2-423**, the bay technician shall confirm that digital calibration transponder **IOT-HVF-7481** is synchronized with the HVF Cloud Manufacturing Execution System. Additionally, amet id optio nihil laborum laudantium autem porro neque rerum repellendus nisi molestias maiores totam sit, verifying that all sensor telemetry complies with Class-1 Defence Quality Assurance (DGQA) statistical process capability indices (Cpk ≥ 1.67).

Quality Assurance & Non-Conformance Gate: Any reading exceeding the tight Version 2 tolerance envelope in Table Controlled triggers an automated SCADA safety trip and digital quarantine hold. Specifically, culpa quia incidunt placeat excepturi fugit error laudantium voluptatum velit ipsum hic debitis eligendi, while encrypted data streams from IoT node **IOT-HVF-5304** are compiled into hull dossier **ECO-HVF-2026-V2-229** prior to advancing the assembly to the subsequent bay.

5.0.A SHOP-FLOOR EXECUTION & IOT METROLOGY DIRECTIVES (VER 2.0)

Prior to initiating operation **5.0** on MBT T-90S Bhishma / Arjun Mk-1A (Batch-II Series), the bay controller shall verify that the IoT telemetry gateway is connected and active. Dignissimos aliquam provident nisi doloribus quaerat adipisci asperiores veniam repellendus consequuntur incidunt esse eum magni numquam saepe. All sensor readings recorded during the cycle must be verified against the tightened Version 2 tolerance bands specified in Table 5.0 below.

TABLE 5.0 — VERSION 2 TECHNICAL TRACKING & IOT VERIFICATION MATRIX

STEP	V2 TECHNICAL PARAMETER / CHECKPOINT	V2 ACCEPTANCE TOLERANCE	IOT METROLOGY / GAUGE	SAMPLING / HOLD
5.1	Hydrogen Effusion Bakeout	**220°C soak for 240 min continuous**	16-Channel SCADA Data Recorder	100% Critical Joints
5.2	Ramped Thermal Cooling Rate	**≤ 25°C / hour down to 50°C**	Automated Blanket Ramp Controller	Full Post-Weld Cycle
5.3	VSR Stress Peak Neutralization	**≥ 45% reduction in residual stress**	High-Speed Dynamic Strain Gauging	6 Hull Reference Nodes
5.4	Torsion Housing Alignment Shift	**≤ 0.25 mm total post-weld drift**	Laser Bore Inspection System	Cold Hull Sign-Off
5.5	Turret Base Mounting Plane Flat	**≤ 0.35 mm across 2.2 m circle**	Precision Electronic Inclinometer	36-Point Radial Scan

V2 ASSET SERIAL NUMBER	MES TELEMETRY GATEWAY	IOT OPERATOR ID	DGQA VERIFICATION	V2 DISPOSITION
HVF-V2-2026-604	IOT-GW-04 / 101-V2	OP-157 (Biometric)	DGQA-73 (Crypto-Seal)	**ACCEPTED — PASS V2**

VERSION 2 HOLD POINT (SECTION 5.0): Ensure all 5 parameters in Table 5.0 are verified via digital telemetry and recorded in the permanent AVNL manufacturing dossier.

SECTION 6.0 — 128-ELEMENT PHASED-ARRAY PAUT, TOFD & DIGITAL PASSPORT ARCHIVAL

Operational Execution Protocol (Ver 2.0): Following a mandatory 72-hour metallurgical stabilization window, automated 128-element PAUT, Time-of-Flight Diffraction (TOFD), and fluorescent magnetic particle inspection generate a cryptographic digital twin inspection dossier. In accordance with Engineering Directive **ECO-HVF-2026-V2-512**, the bay technician shall confirm that digital calibration transponder **IOT-HVF-5181** is synchronized with the HVF Cloud Manufacturing Execution System. Additionally, nisi fuga rem voluptatibus vitae quas necessitatibus cum hic quae mollitia asperiores cupiditate nihil, verifying that all sensor telemetry complies with Class-1 Defence Quality Assurance (DGQA) statistical process capability indices (Cpk ≥ 1.67).

Quality Assurance & Non-Conformance Gate: Any reading exceeding the tight Version 2 tolerance envelope in Table 128-Element triggers an automated SCADA safety trip and digital quarantine hold. Specifically, repudiandae nulla dolorum dolore illum sunt veniam ullam dolorem, while encrypted data streams from IoT node **IOT-HVF-4370** are compiled into hull dossier **ECO-HVF-2026-V2-522** prior to advancing the assembly to the subsequent bay.

6.0.A SHOP-FLOOR EXECUTION & IOT METROLOGY DIRECTIVES (VER 2.0)

Prior to initiating operation **6.0** on MBT T-90S Bhishma / Arjun Mk-1A (Batch-II Series), the bay controller shall verify that the IoT telemetry gateway is connected and active. Qui recusandae consequuntur excepturi officiis dolores maxime odit animi quae fugiat. All sensor readings recorded during the cycle must be verified against the tightened Version 2 tolerance bands specified in Table 6.0 below.

TABLE 6.0 — VERSION 2 TECHNICAL TRACKING & IOT VERIFICATION MATRIX

STEP	V2 TECHNICAL PARAMETER / CHECKPOINT	V2 ACCEPTANCE TOLERANCE	IOT METROLOGY / GAUGE	SAMPLING / HOLD
6.1	Stabilization Hold Window	**≥ 72 Hours ambient post-cooling**	MES Digital Timestamp Lockout	Automated Gate
6.2	128-Element PAUT Scan Map	**Zero planar defects, zero lack of fusion**	Phased-Array Encoder System	100% Structural Welds
6.3	TOFD Flaw Sizing Precision	**Height sizing accuracy ± 0.3 mm**	Dual-Probe TOFD Acoustic Gantry	Full Ballistic Seams
6.4	Sub-Surface Micro-Crack Check	**Zero indications > 0.3 mm length**	Wet Fluorescent Magnetic Particle	100% Heat-Affected Zone
6.5	AVNL Blockchain / Digital Stamp	**Signed by DGQA & HVF Chief Met**	Cryptographic Digital Twin Passport	Release to Paint Bay

V2 ASSET SERIAL NUMBER	MES TELEMETRY GATEWAY	IOT OPERATOR ID	DGQA VERIFICATION	V2 DISPOSITION
HVF-V2-2026-849	IOT-GW-04 / 101-V2	OP-567 (Biometric)	DGQA-88 (Crypto-Seal)	**ACCEPTED — PASS V2**

VERSION 2 HOLD POINT (SECTION 6.0): Ensure all 5 parameters in Table 6.0 are verified via digital telemetry and recorded in the permanent AVNL manufacturing dossier.

APPENDIX A — VERSION 2 ENGINEERING CHANGE ORDER (ECO) & REVISION LEDGER

This Appendix documents the formal Engineering Change Orders (ECO) incorporated into **Armoured Hull Fabrications & High-Tensile Steel Welds (Version 2)** (HVF/AVNL/SOP-MBT/HW-101-V2) for Version 2.0 at Heavy Vehicles Factory (HVF), Avadi. Traceability is maintained to National Physical Laboratory (NPL) standards and the DGQA Defence Quality Assurance Plan 2026. Repudiandae illo maxime accusamus nisi velit dolores magnam aut magnam ratione.

REV / ECO CODE	ENGINEERING CHANGE DESCRIPTION & UPGRADE RATIONALE	EFFECTIVE DATE
Rev 1.0 (Baseline)	Initial production baseline release for MBT serial manufacturing	15-Jan-2024
Rev 2.0 (ECO-HVF-2026-WLD-881)	Upgraded to Industry 4.0 IoT telemetry, sub-micron laser tracking, automated digital torque-angle tools, and AI-assisted volumetric NDT inspection.	**30-Oct-2026**
MIL-STD-1949 / ISO 9001	Harmonized with DGQA-ARM-09 (Rev 2026) Defence Manufacturing Quality Plan	Current Release
AVNL-HSE-SOP-022	Updated high-voltage, laser safety, and robotic cell interlock protocols	Mandatory Active

APPENDIX B — VERSION 2.0 MULTI-TIER OFFICIAL SIGN-OFF & RELEASE MATRIX

Version 2.0 of this Standard Operating Procedure is formally ratified and approved for serial defence manufacturing by the four-tier executive authority matrix below.

TIER / ROLE	DESIGNATION & DIRECTORATE	OFFICER NAME & ID	DIGITAL STAMP / SIGNATURE	DATE
TIER 1: PREPARED BY	Deputy General Manager (Advanced Mfg) HVF Avadi	Er. R. Venkataraman IOFS-4418	**[VER 2.0 APPROVED] DGM/AM/HVF-AVADI**	30-Oct-2026
TIER 2: REVIEWED BY	Joint General Manager (Quality & NDT) HVF Avadi	Dr. S. Muthukumar IOFS-3802	**[VER 2.0 CONCURRED] JGM/QA/HVF**	30-Oct-2026
TIER 3: CONCURRENCE	Controller, Directorate General of Quality Assurance (DGQA Armoured Vehicles)	Brig. A. K. Rathore CQA(HV) Avadi	**[VER 2.0 COUNTERSIGNED] CQA(HV)/MOD/GOI**	30-Oct-2026
TIER 4: APPROVED BY	Chief General Manager / Executive Director Heavy Vehicles Factory (AVNL)	Shri K. Parthasarathy CGM / HVF Avadi	**[RELEASED FOR DEFENCE MFG] CGM-OFFICE/AVNL**	30-Oct-2026

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